ANALYTICAL CUMULATIVE EXAM

October 21, 2005

The essential principle of time-of-flight (TOF) mass spectrometry is that if ions of different mass are given the same kinetic energy, they will acquire different velocities and will therefore have a time of flight through the spectrometer which is mass dependent. A diagram of the spectrometer is attached.

Sample molecules are ionized by electron impact, using an electron beam pulsed for 1 microsecond. Immediately following this ionization pulse, the first accelerating grid is given a pulse of negative charge which starts the positive molecular ions through the first stages of acceleration. The molecular ions reach a drift energy, (typically 2700 electron volts) in a distance of less than 2 cm. Therefore, each molecular ion has a kinetic energy imparted to it of eV, where e is the charge and V is the electrical potential through which the ion passes. The kinetic energy of the same molecular ion can also be given by $1/2\ mv^2$, where m is the mass of the ion, and v is the velocity it achieves.

1. (35 points) Derive the equation for the time needed for a particular ion to pass through the spectrometer.

$$t = L\sqrt{\left(\frac{m}{e}\right)\left(\frac{1}{2V}\right)}$$

where L is the drift path distance (in cm) through the spectrometer and t is the drift time (in μs).

2. (10 points) If the drift length is 100 cm, and the accelerating voltage is 2000 volts, how long would it take an molecular ion with a m/e of 100 to travel through the 100 cm drift path?

3. (20 points) For a drift length of 100 cm, what is the difference in arrival time between ions of m/e = 44 and m/e = 43 when the accelerating voltage is 2800 volts.

4. (10 points) Assuming the same set of parameters in question 3, and that the heaviest ion is m/e = 44, how many times in 1 second could a mass spectra be repeated.

5. (25 points) A mass spectra of an alkane has a molecular ion at 72 m/e which was to small to be used as the base peak. The other peaks along with the relative intensities of each peak is given below.

Mass spect data for a base peak at 57.

m/e	Relative intensity
57	100
41	41.5
29	38.5
27	15.7

Find the empirical formula and the structure of the alkane.
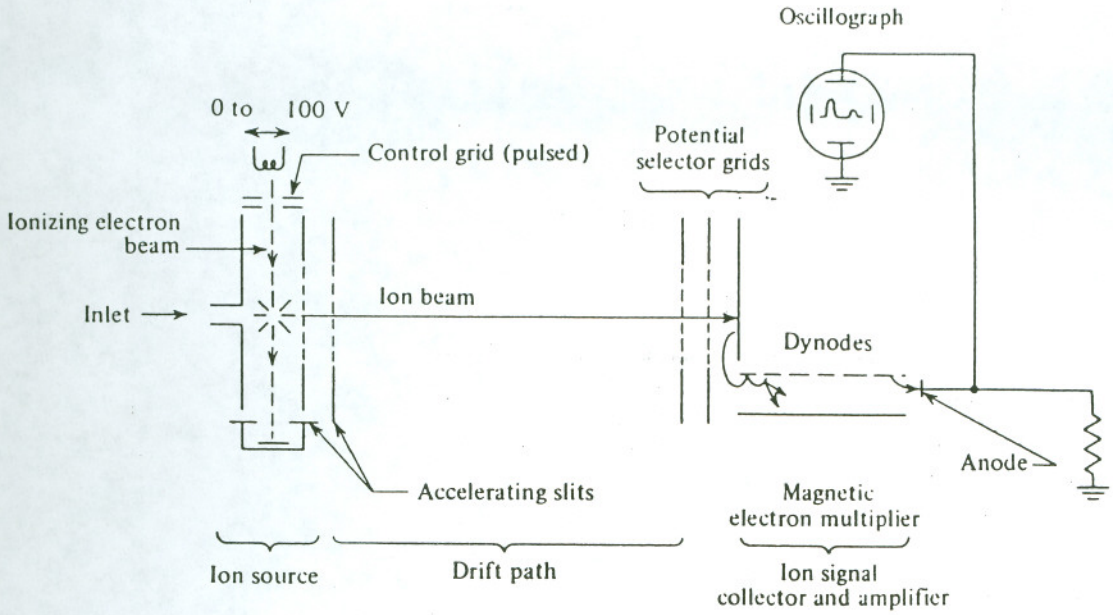
Diagram of TOF Mass spectrometer